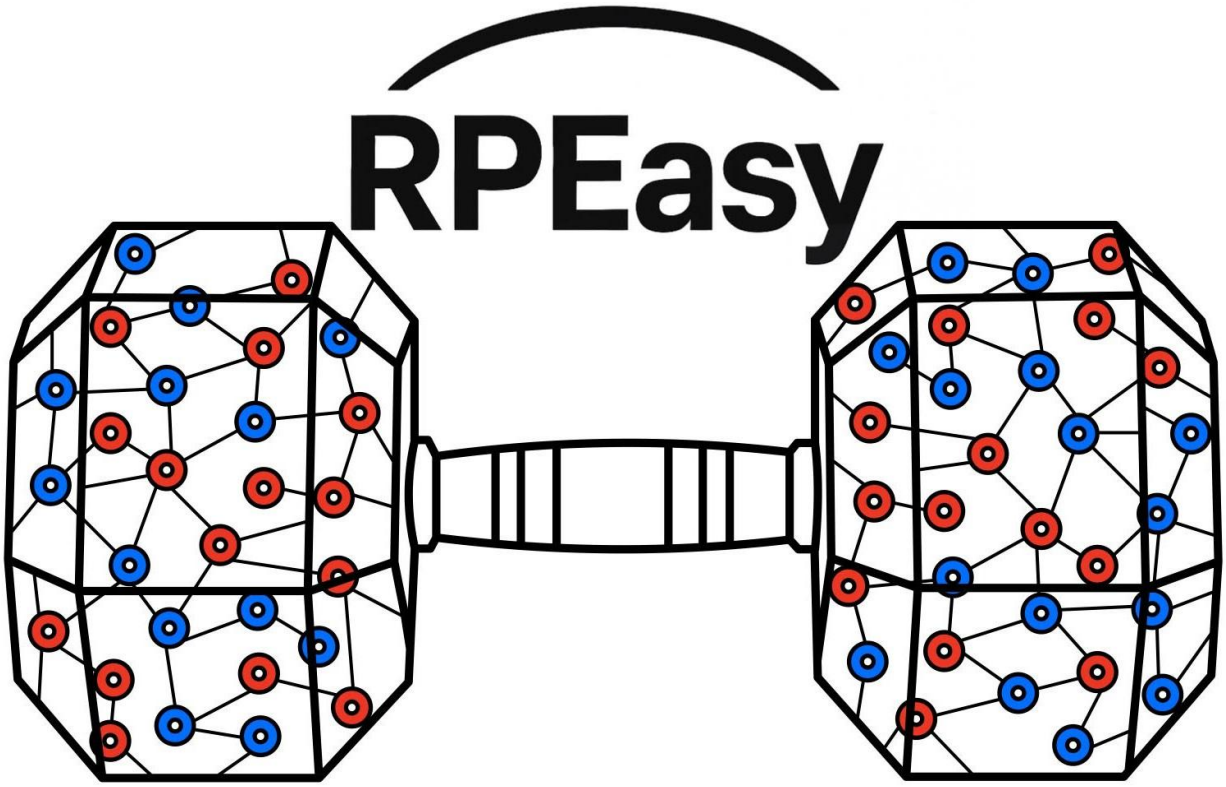
ELEC 490 / 498 Proposal

AI for Weightlifting Athlete



Submitted by Group 53

Group members:

Ajmal Niazi, Luke Hunker, Avedis Boudakian, Rhys Blakeney

Faculty supervisor: Karen Rudie

Executive Summary

The Rate of Perceived Exertion (RPE) during a weightlifting set is a crucial metric to determine an athlete's training intensity, monitor fatigue, and guide progression. The objective of this project is to create an AI-based tool that can predict an athlete's RPE using a video of them squatting, bench pressing, or deadlifting. The video can be broken down into components such as bar speed, facial strain, rep quality, tempo change, and breathing patterns. These components will then be processed independently by a corresponding model, which will then be combined to calculate a final RPE prediction.

During development, we will start with data collection. This will consist of collecting videos of group members doing different lifts at different RPEs. Once enough data has been collected, we will begin the initial development of our computer vision and RPE prediction models. After initial development is complete and we have a minimum viable product, we will iterate to improve the models' accuracy metrics according to our performance specifications.

Evaluation of the component models' performance will occur during both initial training and the optimization stage. The performance of the computer vision components can be manually cross-referenced with input data. Additionally, prediction model accuracy can be tested against a testing dataset. The project can move on from the optimization stage of development once the complete model system meets all our performance specifications.

The required resources for this project include various software components, video recording equipment, and computation time and power. Open-source software exists for our software components, leaving our budget available to use for the recording equipment and cloud GPU resources. Overall, the project is expected to stay within the allotted $600 budget.

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Introduction

The Rate of Perceived Exertion (RPE) is a subjective rating scale that reflects how physically demanding a set feels to a lifter. RPE is a crucial metric when weightlifting, as it helps determine an athlete's training intensity, monitor fatigue, and guide progression.

Two common problems that all weightlifting athletes come across during their training are undertraining and overtraining [7] [8]. A low RPE reveals that an athlete is undertraining and could have performed additional repetitions, creating a higher muscle growth stimulus. On the other end, a high RPE can be advantageous to determine if an athlete is overtraining. Typically, the higher the RPE, the more muscle growth stimulus [9]. However, if an athlete is constantly achieving a high RPE but is not making weightlifting progress (achieving more reps than the previous session while using the same weight), this is a sign that there is too much accumulated muscle fatigue, i.e., overtraining [8].

Developing an AI-based tool that can predict an athlete's RPE (using physical attributes such as bar speed, facial strain, and form breakdown—taken from video) can help athletes determine if they are undertraining or overtraining. New lifters will more commonly experience undertraining as the discomfort of the set causes them to stop prematurely. An AI tool can let beginners know that they had more reps left in a set, which can help encourage them to achieve a higher RPE. Advanced lifters, on the other hand, have the opposite issue and may be achieving too high of an RPE, causing them to overtrain. If the athlete's total weekly volume for an exercise is high, achieving a high RPE every lift may cause more risk than reward, as the muscle may not recover in time for the next session. The AI tool can help advanced lifters by detecting a pattern of high RPE and letting them know to slightly lower the RPE per set or to lower the total weekly volume for the target muscle group.

Currently, there is no product on the market that can help athletes determine training their effectiveness and reduce fatigue by assessing their RPE. This tool would be useful for

beginner and advanced as an unbiased perspective on their RPE can provide important training insights.

The objective of this project is to create a computer vision algorithm that will take video input of the user squatting, bench pressing, or deadlifting and will output the RPE of the set. Using bar speed, facial strain, rep quality, tempo change, and breathing patterns, the algorithm will calculate the RPE. The RPE can then be compared to training frequency and progress to help determine if the athlete is undertraining or overtraining.

The scope of this project includes a final version of a computer vision algorithm capable of taking video input and extracting components such as bar speed, facial strain, rep quality, tempo change, and breathing patterns. When these components are assessed together, they can determine the RPE of a set purely from the visuals. Each extracted component would be processed independently by a corresponding model designed specifically for it. These models would then return their own individual outputs/ predictions, which will then be combined and processed to output one final RPE calculation.

Functional Description

General Description

The objective of this project is to design an AI-based tool that, using computer-vision and machine learning concepts, can predict an athlete's RPE through captured video of training sessions. The system aims to provide automated, data-backed RPE feedback without the need for manual input.

The tool focuses on three fundamental compound lifts (squat, bench press, and deadlift) which will form the basis of video data collection and model training. Using video input, the primary performance indicators used to train the model include bar speed, facial expression/ strain, lifting form, and breathing patterns.

The product will be implemented as a Python based application, designed for integration with standard cameras or smartphones, and will be easily deployable onto local machines and devices with graphics processing capabilities.

Model Specifications

To produce a comprehensive AI-based tool capable of analyzing video feeds and estimating RPE, the final product will require the combination of multiple models and data sources. Each model will process a different aspect of the athlete's video (bar speed, lifting form, facial strain, etc.) and output an individual prediction. These results will then be combined and processed to reach a final consensus on the RPE measurement of that exercise.

The reasoning behind this approach is to reduce computational load and ensure that each model can focus on analyzing its designated feature or physical attribute more effectively than a single, unified model could.

Facial strain + breathing patterns:

When looking specifically at human micro-expressions and breathing detection, pre-existing open-source models such as MediaPipe or OpenFace are readily available and highly effective [4] [5]. These frameworks can detect faces in the video feed, establish key landmark points on the face, and perform deep facial analysis. Some beneficial capabilities of these models in this use case are detecting emotions, facial color, head pose, and Facial Action Units (AU) [6].

Body Posture and Joint Angles:

To assess form and posture, models such as MMPose and MediaPipe can be integrated to detect full body key points and estimate joint positions. These pose estimation frameworks can detect the individual in the frame, generate a bounding box, identify the joints within, and connect landmarks together. This will result in a full-fledged 2D, or 3D skeletal representation of the athlete available to be evaluated on a variety of pose metrics [3] [5].

Bar Speed and Weight detection:

Finally, when tracking the movement of the weights and bars used during these exercises, a simple object detection model such as YOLOv8 or OpenCV can be implemented. These models can easily identify weightlifting tools via attributes like size, colour, or brand markings and track them throughout the video feed [1] [2]. From this motion tracking data, the system will then compute relevant metrics such as bar velocity, acceleration, and range of motion, all of which are useful indicators of athlete fatigue.

Interface Specifications

The AI-based tool will consist of multiple integrated machine learning models that will process video feed data, extract relevant features, and reach an RPE prediction. The following table defines how data will flow between components and some of the expected tolerances.

Components	**Description**	**Data type**	**Tolerance**
Camera Input	Captures video feed data of athlete performing the exercise	RGB video stream (30-60 fps 1080p)	Expected stable frame rate (± 2 frames)
Preprocessing	Video feed frames are extracted, resized, flattened, transformed as necessary for model prediction	Pixel arrays	Pixel normalization ±0.01
Prediction Models	Each model outputs structured data and prediction result	Data structure, array, JSON	Latency ≤100 ms per frame
RPE prediction module	Combines specialized model results using ML model or weighted averages to output predicted RPE	RPE digit (float or int)	Prediction variance ≤5%
User Interface	Final RPE result displayed to athlete and logged	Text output, visual output	Display latency ≤0.5 s

Performance Specifications

The system will be designed to meet the following performance targets under regular testing conditions (one athlete, and proper lighting and video quality).

1. RPE Prediction Accuracy, ±0.5 for a 1-10 RPE scale
2. Pose Detection Accuracy, ≥90% accurate detection of body posture and joint positions
3. Facial Detection Accuracy, ≥90% accurate face detection and strain analysis
4. Bar speed and equipment tracking, ≤3% deviation from measured velocity, acceleration

Risks and safety hazards

When collecting data and making predictions of RPE for the three fundamental exercises (squat, bench press, and deadlift), there is some risk involved when handling heavy weights and safety concerns that must be taken into consideration when designing our product.

An important safety risk to consider is impact of an inaccurate model prediction of RPE being displayed to the user. For example, if the weightlifter receives an inaccurate RPE prediction that was lower than they expected, the lifter may feel inclined to raise the weight for the next exercise when they were already nearing their maximum capacity. This means on the following set there is a higher risk of physical injury that could pose a health and safety risk for the user.

Another safety consideration would be incorrect data collection of attributes such as bar speed or exercise form. If these parameters are shown to the user, it could cause them to drastically change the way they perform the exercise, straying away from what is comfortable for them and pushing them to obtain a better reading. As a result, their form may be strained or they may try and complete the exercises too quickly, causing injury.

Design and Production

The critical path of the design process for this project consists of 4 main milestones: data collection, initial model training, model optimization, and final deployment. Each of these steps has unique challenges and risks that require adequate planning and mitigation. Additionally, each milestone has criteria that must be achieved for them to be considered complete, along with testing procedures to assess each criterion.

The first milestone that we must reach is creating a dataset for training and testing. We will create a dataset by recording videos of ourselves doing each of the three lifts at different intensities and report our RPE. Each member of the group will be needed to perform many sets of each exercise at different RPEs in order to generate a sufficient dataset. Before data collection can begin, we must obtain permission from the General Research Ethics Board to use ourselves as research participants. In the event that we do not receive permission, we will have to source a dataset from information available online.

Initial model training will be the milestone where we will have created our minimum viable product. This will consist of the initial training of each of the models, as well as the model for combining the results from each model to produce a final calculation. We will be able to work on models and processing steps in parallel by assigning different group members to specialize in one modular part of the overall design. While this step is crucial to the overall development process, if any specific model is taking a disproportionate amount of time or resources, the scope of the project can be reduced to exclude certain models while still ensuring a viable final product. This step is complete when we can give the models a new input and receive an estimated RPE as an output.

Model optimization will be the stage of development where we will dial in parameters of the different models to increase the accuracy. This step will conclude once the model meets our performance specifications. Depending on initial model performance and its rate of improvement after iteration, this step could take considerable time before performance

specifications are met. Performance specifications may be modified during this step if they are unfeasible given our time and resource constraints.

Final deployment is the final step of the project's development in which we will prepare the models and UI for presentation. The scope and depth of this step is flexible, dependent on the timeliness of the completion of previous steps. This step will be where we finish our work on the project, polishing the UI as much as time allows as well as making any final tweaks to our end product.

Testing and Evaluation

Each of our RPE prediction models rely on pre-processing of video to isolate one specific parameter for analysis. During the initial training stage of the project, the accuracy of the pre-processing will be evaluated against our performance specifications. For facial recognition and pose detection, our training data will be visually cross-referenced with the outputs of the computer vision models. Bar speed detection can also be manually calculated and cross-referenced with model outputs.

To test the performance of our trained model, we will divide our dataset into a training set and a testing set. We expect to use a standard division of 80% of data for the training set, and 20% of the data for the testing set. Once trained, a model's performance will be evaluated on its accuracy when predicting the RPE of a new datapoint from the testing set. Each model will be tested individually, and the combined score of the models will also be evaluated to optimize the calculation of the combined RPE prediction. This testing will start after the initial model training and will continue throughout the model optimization stage where we will iterate on models according to their test scores. While maximum accuracy is the goal, a prediction accuracy of +/- 0.5 points on the RPE scale will be considered a success, especially given the subjectivity of score perception and reporting.

Resource requirements

Scheduling

To complete the project successfully, our team will use different resources at different stages of development. The following table describes the project milestones and resources required at every level.

Milestone	**Resources Required**
Data Collection	Queen's University weight room facilities, cameras (HD video capture), tripods, consent forms and GREB approval, student participants
Initial Model Training	High-performance computing resources (GPU-enabled workstations or access to Queen's Compute Cluster), Python ML libraries (TensorFlow, PyTorch), open-source repos (MediaPipe, OpenFace, MMPose)
Model Optimization and Validation	Additional compute time, access to stored dataset, annotation software for labeling reps and facial strain, statistical software (MATLAB, Python)
Final Deployment & Showcase	Laptop/ desktop with GPU acceleration for demo, projector/ screen for presentation, prepared demo dataset

Material and Other Resources

To properly design, implement, and test the AI-based RPE prediction system, a combination of hardware, software, and physical resources will be required throughout the

project. Most of the tools and technologies required are open-source or available through Queen's University student licenses, minimizing additional project costs.

The primary hardware resources are high-performance computing infrastructure access to train and process machine learning models and video data. This will preferably be GPU enabled workstations or access to the Centre for Advanced Computing (CAC), which will make it possible to efficiently process large video datasets and deep learning libraries such as TensorFlow or PyTorch. There will also be additional standard desktop computers and laptops used for software development, testing of code, and model deployment.

For data collection, the team would require cameras that are capable of recording at least 1080p and 60 frames per second to capture accurate motion, bar speed, and facial tension. Tripods and appropriate lighting equipment will also be required to ensure video quality is consistent throughout sessions. Recordings will be done in the Queen's University gymnasium facilities pending approval by the General Research Ethics Board (GREB). The videos will serve as validation and training data for the AI model.

The software requirements for this project include primarily machine learning and computer vision libraries like TensorFlow, PyTorch, OpenCV, MediaPipe, OpenFace, and MMPose. The code will be developed on PyCharm and Jupyter Notebook with the version control hosted on GitHub using the free student license. MATLAB or Python libraries such as NumPy and pandas will be utilized for statistical analysis and data processing. All packages selected are open-source or are available under Queen's academic licensing arrangements.

Physical equipment costs like camera and tripod estimates range from $200–$400 depending on if materials are borrowed or rented from Queen's University. If GPU or compute cluster availability on campus is limited, the team may allocate up to $100–$200 in cloud computing credits (e.g., AWS, Google Colab Pro, or Microsoft Azure) to ensure continued progress during model training.

Finally, the group anticipates using university provided lab spaces such as the Bain Lab for testing, debugging, and final demonstrations. These spaces will also be used for integration

testing and preparation of the final project showcase. Overall, the project's resource requirements are realistic and accessible within the existing Queen's infrastructure, with minimal expected additional expenditures.

Conclusion

The proposed project aims to develop an AI-based system that can accurately estimate the RPE of an athlete from video footage of simple compound lifts. Employing sophisticated computer vision and machine learning techniques, the system will analyze key performance indicators such as bar speed, facial strain, form, and breathing patterns to output an objective RPE value. This automated feedback mechanism has the opportunity to help athletes identify undertraining or overtraining, optimize recovery, and improve training efficiency.

Technical sophistication and usability are both evidenced in the development and design approach outlined in this proposal. By using structured milestones (data collection, model training, optimization, and deployment), the group will set up a modular and extensible tool that integrates multiple specialized models into one prediction system. Utilization of open-source libraries such as MediaPipe, OpenFace, MMPose, and YOLOv8, as well as facilities provided by Queen's University such as the Centre for Advanced Computing and lab spaces on campus, ensures the project remains cost-effective and accessible.

Completion of this project will result in a functional proof of concept capable of evaluating athlete exertion with measurable accuracy. Beyond its research value, the tool symbolizes a contribution to the growing intersection of artificial intelligent and exercise science, providing a foundation for further research into athlete tracking, injury prevention, and performance optimization.

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